Software Documentation

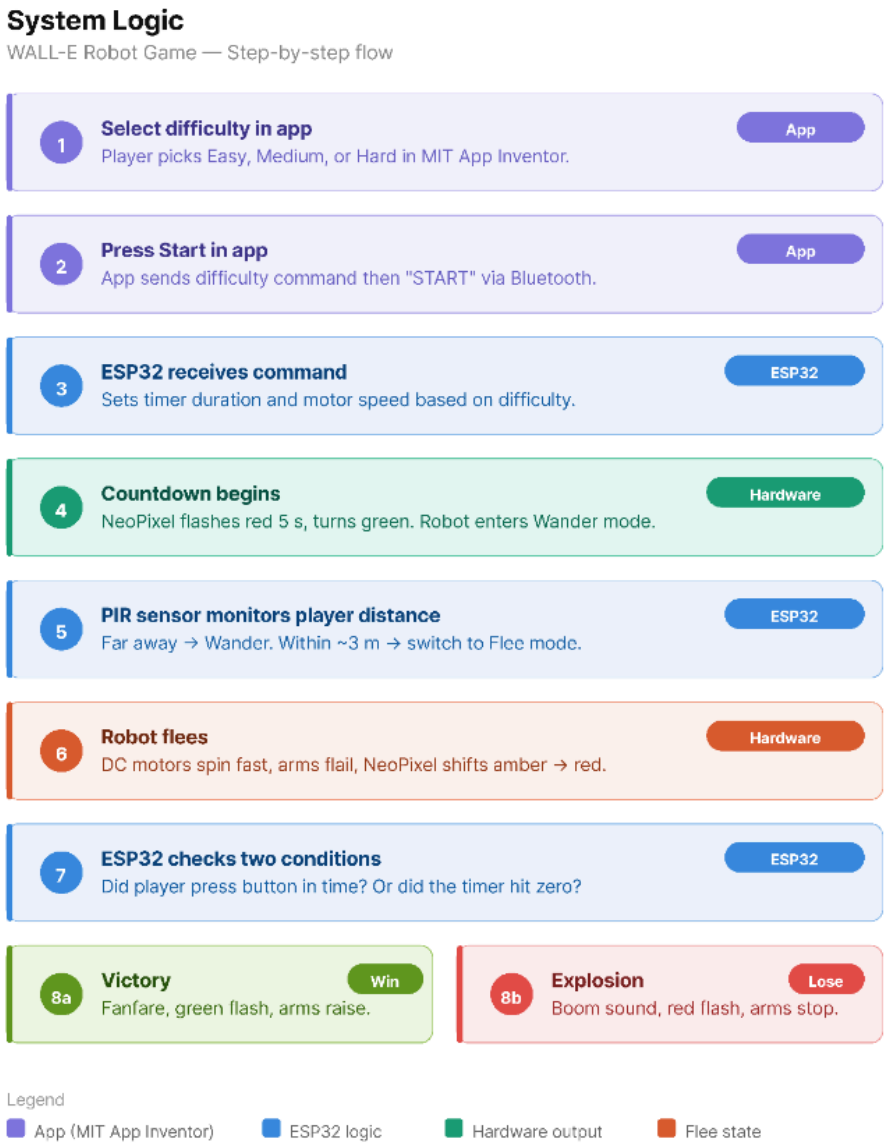
1. Software Overview

- *The software controls a WALL-E robot game where players chase and tag a fleeing robot before a countdown timer expires. It handles Bluetooth communication between the MIT App Inventor phone app and the ESP32, manages game states (Wander, Flee, Victory, Explosion), drives motors, arms, NeoPixel, and buzzer, and reads the PIR and IR sensors, and push button.*

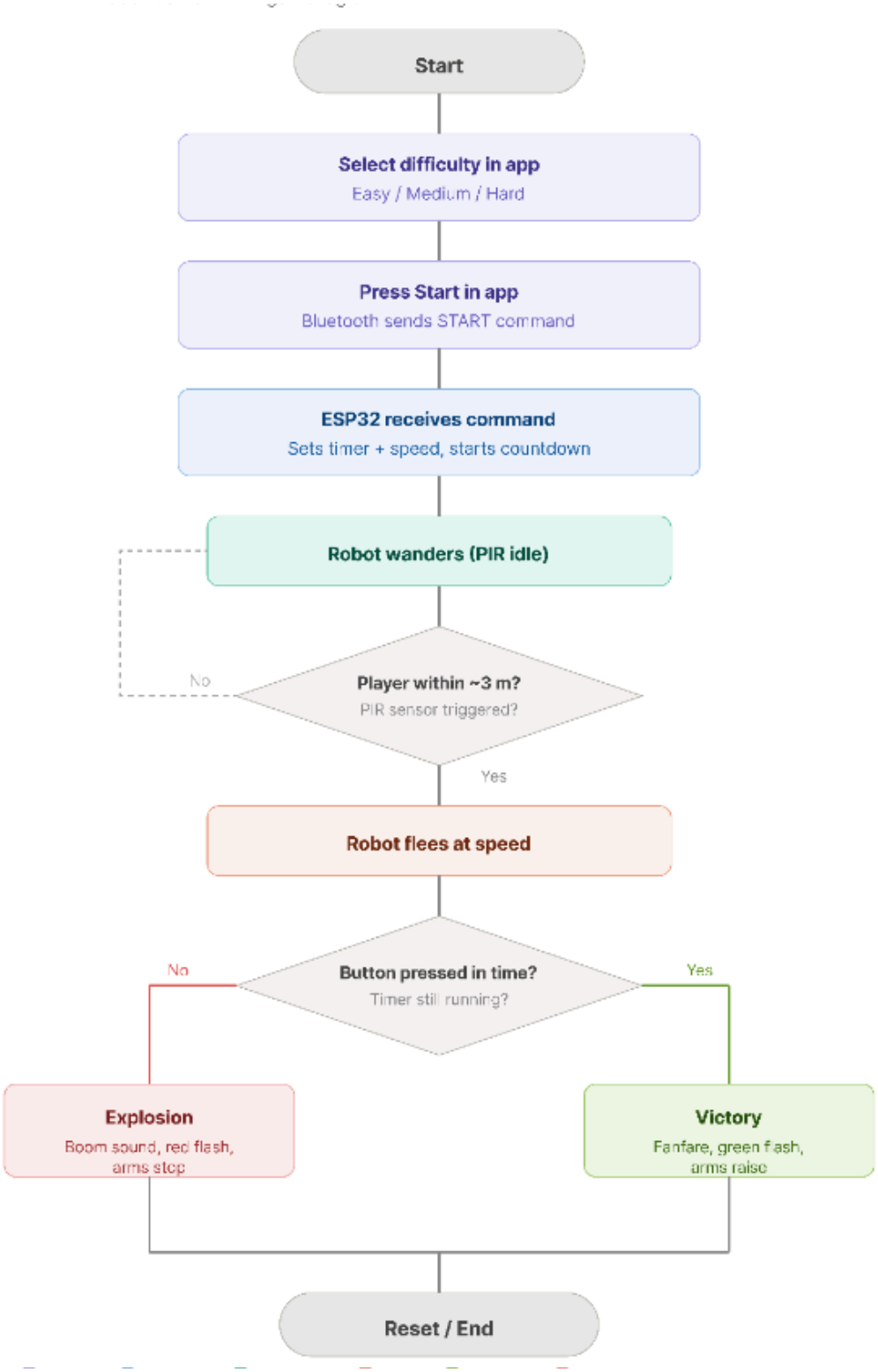
- *MicroPython / Thonny: runs on the ESP32 to control all hardware: motors, sensors, NeoPixel ring, and buzzer. Handles Bluetooth receive, game logic, and state transitions.*

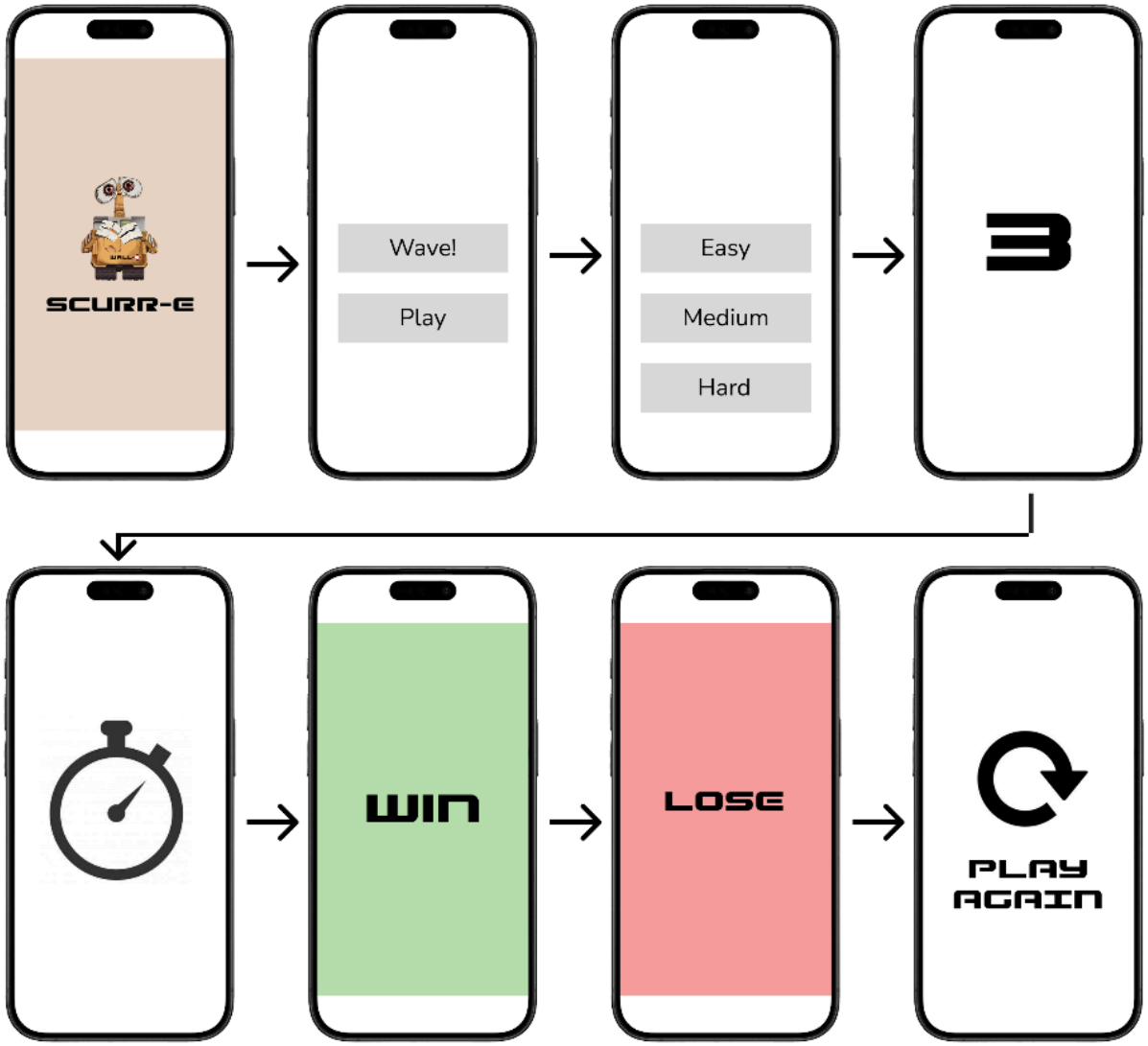
  MIT App Inventor: The app that lets the player select a difficulty level (Easy / Medium / Hard) and press Start or Stop. Sends commands wirelessly to the ESP32 via Bluetooth.

2. System Logic



3. Flowchart


Start
Select difficulty in app
Easy / Medium / Hard
Press Start in app
Bluetooth sends START command
ESP32 receives command
Sets timer + speed, starts countdown
Robot wanders (PIR idle)
No
Player within ~3 m?
PIR sensor triggered?
Yes
Robot flees at speed
No
Button pressed in time?
Timer still running?
Yes
Explosion
Boom sound, red flash, arms stop
Victory
Fanfare, green flash, arms raise
Reset / End

4. App Flow (If MIT App Inventor)

- *App flow rough draft: (NOT FINAL APP DESIGN)*